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Bilinear interpolation theorems and applications ☆

Mieczysław Mastyło

Faculty of Mathematics and Computer Science, Adam Mickiewicz University, and Institute of Mathematics, Polish Academy of Sciences (Poznań branch), Umultowska 87, 61-614 Poznań, Poland

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Abstract

Interpolation of bilinear operators is studied. The celebrated Littlewood mixed norm inequality is used to prove interpolation theorems for bilinear operators defined on couples of c_0-weighted sequence spaces generated by parameters of quasi-concave functions. It is shown that these results can be lifted to a wider class of abstract methods of interpolation. The abstract results are used to prove bilinear interpolation theorems to the setting of Calderón–Lozanovskii spaces. As an application the boundedness of the bilinear Hilbert transform between Orlicz spaces including Zygmund classes is proved.

Keywords: Bilinear operator; The bilinear Hilbert transform; Interpolation functor; Calderón–Lozanovskii space

1. Introduction

Many sophisticated methods of interpolation, mainly within the interpolation theory of linear operators which bear applications in other areas of mathematics, were developed. Some time ago it became clear that the classical bilinear interpolation theorems due to Calderón [8] for the complex method of interpolation and Lions and Peetre [26] for the real method of interpolation are important to several disciplines, including the harmonic analysis and various branches of functional analysis. Their results state that if (X_0, X_1), (Y_0, Y_1) and (Z_0, Z_1) are compatible couples of Banach spaces, and $T_0 : X_0 \times Y_0 \to Z_0$, $T_1 : X_1 \times Y_1 \to Z_1$ are bounded bilinear operators

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E-mail address: mastylo@amu.edu.pl.

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which agrees with a bilinear map T from $(X_0 \cap X_1) \times (Y_0 \cap Y_1) \to Z_0 \cap Z_1$, then for the complex method, T has a bilinear extension from $[X_0, X_1]_\theta \times [Y_0, Y_1]_\theta$ into $[Z_0, Z_1]_\theta$, while for the real method, T has a bilinear extension from $(X_0, X_1)_{\theta,p} \times (Y_0, Y_1)_{\theta,q}$ into $(Z_0, Z_1)_{\theta,r}$ provided that $p, q, r \in [0, \infty]$ satisfy $1/r = 1/p + 1/q - 1$ (see, e.g., [4, Theorem 4.4.1] and [26] or [8] and [36]). Much less is known about bilinear interpolation using other interpolation methods. In the light of the mentioned classical bilinear interpolation theorems the following natural question appears. For which method of interpolation, can one conclude that T has a bounded bilinear map between the associated interpolation spaces? This line of research was considered by the author in [29] for the minimal and maximal methods of interpolation introduced by Aronszajn and Gagliardo [1]. In [29] the author prove abstract bilinear interpolation theorems in those contexts and then use them to derive various more particular theorems such as a bilinear interpolation theorem for Orlicz spaces. It should be noticed here that the minimal and maximal methods lie at the heart of the abstract theory of interpolation (see [7,20,33]) and will be used here as well.

In this paper we consider the mentioned question. The main aim is to prove new abstract bilinear interpolation theorems and show applications. This is motivated by a question about whether some of the results proved in recent years for important bilinear operators between L_p-spaces have extensions to more general function spaces. Examples of such operators are for example the bilinear multipliers. Recall that a bounded measurable function σ on $\mathbb{R}^n \times \mathbb{R}^n$ generates to a bilinear operator W_σ defined by

$$W_\sigma(f, g)(x) = \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \sigma(\xi, \eta) \hat{f}(\xi) \hat{g}(\eta) e^{2\pi i \langle x, \xi+\eta \rangle} \, d\xi \, d\eta, \quad x \in \mathbb{R}^n$$

for every f, g in the Schwartz space $\mathcal{S}(\mathbb{R}^n)$, where $\langle \cdot, \cdot \rangle$ denotes the inner product in $\mathbb{R}^n$. In this context σ is called the symbol of W_σ. The study of such bilinear multiplier operators was initiated by Coifman and Meyer in [10], where it is proved that if $1 < p, q < \infty$, $1/r = 1/p + 1/q$ and the function σ on $\mathbb{R}^n \times \mathbb{R}^n$ satisfies

$$\left| \partial_\xi^\alpha \partial_\eta^\beta \sigma(\xi, \eta) \right| \leqslant C_{\alpha,\beta} \big(|\xi| + |\eta| \big)^{-|\alpha|-|\beta|}$$

for sufficiently large multi-indices α and β, then W_σ extends to a bilinear operator from $L_p(\mathbb{R}^n) \times L_q(\mathbb{R}^n)$ into $L_{r,\infty}(\mathbb{R}^n)$ whenever $r > 1$. Here as usual $L_{r,\infty}(\mathbb{R}^n)$ denotes the weak L_r space of Marcinkiewicz. This result was later extended to the range $1 > r > 1/2$ by Grafakos and Torres [18] and Kenig and Stein [23]. The bilinear multipliers of Marcinkiewicz type were studied by Grafakos and Kalton [15]. The first significant boundedness results concerning non-smooth symbols were proved by Lacey and Thiele [24,25] who established that W_σ with $\sigma(\xi, \eta) = \operatorname{sign}(\xi + \alpha\eta)$, $\alpha \in \mathbb{R} \setminus \{0, 1\}$ has a bounded extension from $L_p(\mathbb{R}^n) \times L_q(\mathbb{R}^n)$ to $L_r(\mathbb{R}^n)$ if $2/3 < r < \infty$, $1 < p, q \leqslant \infty$, and $1/r = 1/p + 1/q$. Extensions of this result was subsequently obtained by Gilbert and Nahmod [14]. We refer to an interesting article by Blasco [5] which reviews the properties of the classes of bilinear multiplier operators acting on Lebesgue spaces and provides also some new results.

Although boundedness of various bilinear operators on products of L_p-spaces have been studied extensively, surprisingly little is known about boundedness of these operators on products of more general function spaces, e.g., Orlicz spaces. Unfortunately, the techniques used in the case of L_p-spaces do not always apply directly in a more general setting, and thus more involved abstract methods are required. Therefore we are interested in how far known results on boundedness of classical bilinear operators between L_p-spaces can be extended to more general class of spaces. To study this problem we apply interpolation methods. The main source of inspiration for this is a well-known fact that Orlicz spaces can be generated by the Calderón–Lozanovskii construction, thus bilinear interpolation theorems for these spaces could be applied to a large variety of important bilinear operators between L_p-spaces, in particular to the bilinear Hilbert transform. It should be mentioned here that complex method of interpolation is not helpful here. The main reason is that if $p_0 \geqslant 1$, $p_1 \geqslant 1$ and $0 < \theta < 1$, then the complex interpolation space $[L_{p_0}, L_{p_1}]_\theta = L_p$, where $1/p = (1-\theta)/p_0 + \theta/p_1$ (see [4, Theorem 5.1.1]). Thus the complex interpolation spaces between L_p-spaces are again L_p-spaces. Since interpolation of bilinear operators by general real methods are known (see [3,17]), this explains why we study different than complex or real methods of interpolation which behave well for bilinear operators. The paper is organized as follows. In the next section we introduce the necessary notation and discuss some preliminary results on convolution series generated by bilinear operators from $c_0 \times c_0$ into (quasi)-Banach spaces Y. Using the celebrated Littlewood's (ℓ_1, ℓ_2)-mixed norm inequality we prove that special convolution series generated by bounded double sequences form 2-weakly summable sequences provided Y is a Banach space. We also show summability of the mentioned variant type of series in the case when Y is a maximal p-convex quasi-Banach lattice. The key here is the 2-dimensional Khinchin's inequality.

The core of the paper is Section 3. There we prove interpolation theorems for bilinear operators from products of weighted c_0 spaces generated by quasi-concave functions into any Banach couple. Based on these results, we provide a general framework which allows us to derive powerful extensions to a large class of abstract methods of interpolation. We prove an abstract optimal bilinear interpolation theorem for operators from Banach spaces generated by the Gustavsson–Peetre method defined in [19]. We show that under some conditions on parameters a general bilinear interpolation theorem holds, where the range space for interpolated bilinear operators is generated by Ovchinnikov's upper method of interpolation introduced in [32]. In this class of parameters the result is also optimal. We apply our results in the setting of Banach lattices generated by the Calderón–Lozanovskii spaces that include, in particular, Orlicz spaces.

In Section 4, we apply our new bilinear interpolation theorems to prove the boundedness of the bilinear Hilbert transform on products of Zygmund classes.

2. Preliminary results

We begin with some notation and conventions. Let $(\Omega, \mu) = (\Omega, \Sigma, \mu)$ be a complete σ-finite measure space and let $L^0(\mu)$ denote the space of equivalence classes of real valued measurable functions on Ω, equipped with the topology of convergence (in the measure μ) on sets of finite measure. By a *quasi-Banach lattice* on Ω, we shall mean a quasi-Banach space X which is a subspace of $L^0(\mu)$ such that there exists $u \in X$ with $u > 0$ and if $|x| \leqslant |y|$ a.e., where $y \in X$ and $x \in L^0(\mu)$, then $x \in X$ and $\|x\|_X \leqslant \|y\|_X$. A quasi-Banach lattice X is said to be *maximal* if its unit ball $B_X := \{x;\ \|x\|_X \leqslant 1\}$ is a closed subset in $L^0(\mu)$. If X is a quasi-Banach lattice on (Ω, μ) and $w \in L^0(\mu)$ with $w > 0$ a.e., we define the weighted quasi-Banach lattice $X(w)$ by $\|x\|_{X(w)} = \|xw\|_X$. Recall that an operator T defined on a product of two quasi-Banach spaces $X \times Y$ and taking values in another quasi-Banach space Z is called a *bounded bilinear* (*bilinear* for short) if it is linear in each of the two variables, and bounded, i.e., there is a constant C such that for all $x \in X$ and $y \in Y$, we have

$$\left\|T(x, y)\right\|_Z \leqslant C\|x\|_X\|y\|_Y.$$

The smallest C so that the above inequality holds for all $x \in X$ and $y \in Y$ is called the norm of T and will be denoted by $\|T\|_{X\times Y\to Z}$. We will deal with vector-valued Banach sequence spaces. Let X be a Banach space and $1 \leqslant p < \infty$. The sequence $\{x_n\}_{n\in\mathbb{Z}}$ is said to be *weakly p-summable* if the scalar sequences $\{x^*(x_n)\}$ are in ℓ_p for every $x^* \in X^*$. The space of all weakly p-summing sequences in X is denoted by $\ell_p^w(X)$. It is a Banach space equipped with the norm

$$\big\|\{x_n\}\big\|_{\ell_p^w(X)} := \sup\left\{\left(\sum_{n\in\mathbb{Z}}\big|x^*(x_n)\big|^p\right)^{1/p};\ \|x^*\|_{X^*} \leqslant 1\right\}.$$

We begin with proving a lemma on series which will be used later. In the proof we will use Littlewood's (ℓ_1, ℓ_2)-mixed norm inequality (see [27], for details and history we refer to [6]). We state it in a form convenient for us: there exists a constant $\kappa_L \leqslant \sqrt{2}$ such that for all arrays $\beta = (\beta(m,n))_{m,n\in\mathbb{Z}}$, we have

$$\sum_m\left(\sum_n\big|\beta(m,n)\big|^2\right)^{1/2} \leqslant \kappa_L \sup\left\|\sum_{m\in S, n\in T}\beta(m,n)\varepsilon_m\otimes\varepsilon_n\right\|_{L_\infty([0,1]\times[0,1])},$$

where the supremum is taken over all finite subsets $S, T \subset \mathbb{Z}$ and $\{\varepsilon_k\}_{k\in\mathbb{Z}}$ is a sequence of independent Bernoulli random variables on $[0,1]$. From now on we shall assume in the paper that a Banach sequence spaces c_0 and ℓ_p are modeled on the set of all integers $\mathbb{Z}$.

Lemma 2.1. *Let Y be a Banach space and let $T : c_0 \times c_0 \to Y$ be a bilinear operator with the norm $\|T\|$ and let $\{a_{km}\}_{k,m\in\mathbb{Z}}$ be a bounded double sequence.*

(i) *If $\{a_{km}\}_{k\in\mathbb{Z}} \in c_0$ for each $m \in \mathbb{Z}$, then the series $\sum_{k\in\mathbb{Z}} a_{km}T(e_k, e_{m-k})$ converges in Y for each $m \in \mathbb{Z}$ and*

$$\left\|\sum_{k\in\mathbb{Z}} a_{km}T(e_k, e_{m-k})\right\|_Y \leqslant \|T\| \sup_{k\in\mathbb{Z}}|a_{km}|.$$

(ii) *There exists a constant $\kappa_L \leqslant \sqrt{2}$ such that for any pair of sequences $\xi = \{\xi_n\}$, $\eta = \{\eta_n\}$ in c_0 the sequence $\{y_m\}_{m\in\mathbb{Z}}$ defined by $y_m := \sum_{k\in\mathbb{Z}} a_{km}\xi_k\eta_{m-k}T(e_k, e_{m-k})$ is weakly 2-summable in Y. Moreover, the following estimate holds*

$$\sup_{\|y\|_{Y^*}\leqslant 1}\left(\sum_{m\in\mathbb{Z}}\big|y^*(y_m)\big|^2\right)^{1/2} \leqslant \kappa_L\|T\|\|\xi\|_{c_0}\|\eta\|_{c_0}\sup_{k,m\in\mathbb{Z}}|a_{km}|.$$

Proof. (i). For every $y^* \in Y^*$ with $\|y^*\|_{Y^*} \leqslant 1$, we define a bilinear form $B_{y^*} : c_0 \times c_0 \to \mathbb{K}$ by $B_{y^*} = y^* \circ T$. By [2], we have

$$\sum_{k\in\mathbb{Z}}\big|B_{y^*}(e_k, e_{m-k})\big| \leqslant \|B_{y^*}\|, \quad m \in \mathbb{Z}.$$

This implies that the series $\sum_{k\in\mathbb{Z}} T(e_k, e_{m-k})$ is weakly conditionally convergent in Y. Since $\|B_{y^*}\| \leqslant \|T\|$, it follows that for any finite subset $F \subset \mathbb{Z}$, we have

$$\begin{aligned}\Big\|\sum_{k\in F} a_{km} T(e_k, e_{m-k})\Big\|_Y &\leqslant \sup_{\|y^*\|_{Y^*}\leqslant 1} \sum_{k\in F} \big|a_{km} y^*\big(T(e_k, e_{m-k})\big)\big| \\ &\leqslant \Big(\sup_{\|y^*\|_{Y^*}\leqslant 1} \sum_{k\in\mathbb{Z}} \big|B_{y^*}(e_k, e_{m-k})\big|\Big) \sup_{k\in F} |a_{km}| \\ &\leqslant \|T\| \sup_{k\in F} |a_{km}|\end{aligned}$$

and this completes the proof of (i).

(ii). We may assume that $|a_{km}| \leqslant 1$ for all $(k, m) \in \mathbb{Z}\times\mathbb{Z}$. Let $y^* \in Y^*$ with $\|y^*\|_{Y^*} \leqslant 1$. We define an infinite scalar array $\beta = (\beta(m, n))$ by

$$\beta(m, n) := y^*\big(T(\xi_m e_m, \eta_n e_n)\big), \quad m, n \in \mathbb{Z}.$$

By Minkovski's inequality we obtain

$$\begin{aligned}&\Big(\sum_m \Big|\sum_k a_{km}\xi_k\eta_{m-k} y^*\big(T(e_k, e_{m-k})\big)\Big|^2\Big)^{1/2} \\ &\quad\leqslant \Big(\sum_m\Big(\sum_k \big|\beta(k, m-k)\big|\Big)^2\Big)^{1/2} = \sum_m\Big(\sum_n \big|\beta(m, n)\big|^2\Big)^{1/2}.\end{aligned}$$

It then follows from Littlewood's (ℓ_1, ℓ_2)-mixed norm inequality that

$$\Big(\sum_m \big|y^*(y_m)\big|^2\Big)^{1/2} \leqslant \sum_m\Big(\sum_n \big|\beta(m, n)\big|^2\Big)^{1/2} \leqslant \kappa_L \|T\| \|\xi\|_{c_0} \|\eta\|_{c_0},$$

and this completes the proof. $\square$

In what follows we will need the 2-dimensional Khinchin's inequality (which is a consequence of the classical Khinchin's inequality). It states that for any $0 < p < \infty$ and for any finite double sequence $\{a_{km}\}$ of scalars the following inequalities hold:

$$\begin{aligned}A_p^2\Big(\sum_{k,m} |a_{km}|^2\Big)^{1/2} &\leqslant \Big(\int_0^1\int_0^1 \Big|\sum_{k,m} \varepsilon_k(s)\varepsilon_m(t)a_{km}\Big|^p\, ds\, dt\Big)^{1/p} \\ &\leqslant B_p^2\Big(\sum_{k,m} |a_{km}|^2\Big)^{1/2},\end{aligned}$$

where as usual $\{\varepsilon_k\}_{k\in\mathbb{Z}}$ is the sequence of independent Bernoulli random variables on [0, 1] and A_p and B_p are the constants from Khinchin's inequality.

For formulation of the next result, some additional notation and definitions are required. Given a quasi-Banach lattice X and $0 < p < \infty$, we define the *p-concavification* X_p to be a quasi-Banach lattice of all $x \in L^0(\mu)$ such that $|x|^{1/p} \in X$ equipped with the quasi-norm $\|x\|_{X_p} =$

$\||x|^{1/p}\|^p$. Following [21], X is said to be *p-convex* if there exists a constant $C > 0$ such that for any $x_1, \ldots, x_n \in X$, we have

$$\left\| \left(\sum_{k=1}^{n} |x_k|^p \right)^{1/p} \right\|_X \leqslant C \left(\sum_{k=1}^{n} \|x_k\|_X^p \right)^{1/p}.$$

The least C is denoted by $M^{(p)}(X)$.

Note that if X is p-convex, then for any $x \in X_p$ and $x_1, \ldots, x_n \in X_p$ with $|x| \leqslant \sum_{k=1}^{n} |x_k|$, we have

$$\begin{aligned} \|x\|_{X_p} &\leqslant \left\| \left(\sum_{k=1}^{n} \left(|x_k|^{1/p}\right)^p \right)^{1/p} \right\|_X^p \leqslant M^{(p)}(X)^p \sum_{k=1}^{n} \left\| |x_k|^{1/p} \right\|_X^p \\ &= M^{(p)}(X)^p \sum_{k=1}^{n} \|x_k\|_{X_p}. \end{aligned}$$

This shows that the lattice norm $\| \cdot \|^*$ defined on X_p by

$$\|x\|^* := \inf \left\{ \sum_{k=1}^{n} \|x_k\|_{X_p};\ |x| \leqslant \sum_{k=1}^{n} |x_k|,\ n \in \mathbb{N} \right\}, \quad x \in X_p$$

satisfies

$$M^{(p)}(X)^{-p} \| \cdot \|_{X_p} \leqslant \| \cdot \|^* \leqslant \| \cdot \|_{X_p}.$$

In what follows a quasi-Banach X is said to has nontrivial convexity whenever it is p-convex for some $0 < p < \infty$.

Lemma 2.2. *Let Y be a maximal p-convex quasi-Banach lattice on (Ω, μ) and $T : c_0 \times c_0 \to Y$ a bilinear operator. If a sequence $\{a_{km}\}_{k,m \in \mathbb{Z}}$ is such that $C := \sup_{m \in \mathbb{Z}} (\sum_{k \in \mathbb{Z}} |a_{km}|^2)^{1/2} < \infty$, then the series $\sum_{k \in \mathbb{Z}} a_{km} T(e_k, e_{m-k})$ converges in Y for each $m \in \mathbb{Z}$. Moreover, if for each $m \in \mathbb{Z}$ we define*

$$y_m = \sum_{k \in \mathbb{Z}} a_{km} T(e_k, e_{m-k}), \quad m \in \mathbb{Z},$$

then for any sequence $\{I_n\}_{n \in \mathbb{Z}}$ of disjoint sets $I_n \subset \mathbb{Z}$ and any sequence $\{\delta_m\}_{m \in \mathbb{Z}}$ of real numbers with $|\delta_m| \leqslant 1$ for each $m \in \mathbb{Z}$, we have $(\sum_{n \in \mathbb{Z}} |\sum_{m \in I_n} \delta_m y_m|^2)^{1/2} \in Y$ and

$$\left\| \left(\sum_{n \in \mathbb{Z}} \left| \sum_{m \in I_n} \delta_m y_m \right|^2 \right)^{1/2} \right\|_Y \leqslant C A_p^{-2} M^{(p)}(Y) \|T\|_{c_0 \times c_0 \to Y}.$$

Proof. Fix positive integers M, N. An application of the 2-dimensional Khinchin's inequality implies the pointwise estimates

$$\begin{aligned}
&\Big(\sum_{|k|\leqslant N}\sum_{|m|\leqslant M}\big|T(e_k,e_m)\big|^2\Big)^{1/2}\\
&\leqslant A_p^{-2}\bigg(\int_0^1\int_0^1\Big|\sum_{|k|\leqslant N}\sum_{|m|\leqslant M}\varepsilon_k(s)\varepsilon_m(t)T(e_k,e_m)\Big|^p\,ds\,dt\bigg)^{1/p}\\
&= A_p^{-2}\bigg(\int_0^1\int_0^1\Big|T\Big(\sum_{|k|\leqslant N}\varepsilon_k(s)e_k,\sum_{|m|\leqslant M}\varepsilon_m(t)e_m\Big)\Big|^p\,ds\,dt\bigg)^{1/p}.
\end{aligned}$$

As it was noticed above there exists a lattice norm $\|\cdot\|^*$ on Y_p such that $C_1\|\cdot\|_{Y_p}\leqslant\|\cdot\|^*\leqslant\|\cdot\|_{Y_p}$, where $C_1 := M^{(p)}(Y)^{-p}$. This implies that for every $y\in Y$, we have

$$C_1\|y\|_Y^p = C_1\big\||y|^p\big\|_{Y_p}\leqslant\big\||y|^p\big\|^*\leqslant\|y\|_Y^p.$$

Combining the above inequalities with the fact that $\|\cdot\|^*$ is a lattice, we obtain an estimate

$$\begin{aligned}
&C_1\bigg\|\Big(\sum_{|k|\leqslant N}\sum_{|m|\leqslant M}\big|T(e_k,e_m)\big|^2\Big)^{1/2}\bigg\|_Y^p\\
&\leqslant A_p^{-2p}\int_0^1\int_0^1\bigg\|\Big|T\Big(\sum_{|k|\leqslant N}\varepsilon_k(s)e_k,\sum_{|m|\leqslant M}\varepsilon_m(t)e_m\Big)\Big|^p\bigg\|^*\,ds\,dt\\
&\leqslant A_p^{-2p}\int_0^1\int_0^1\Big\|T\Big(\sum_{|k|\leqslant N}\varepsilon_k(s)e_k,\sum_{|m|\leqslant M}\varepsilon_m(t)e_m\Big)\Big\|_Y^p\,ds\,dt\\
&\leqslant A_p^{-2p}\|T\|_{c_0\times c_0\to Y}^p.
\end{aligned}$$

This gives by maximality of Y that $y := (\sum_{k\in\mathbb{Z}}\sum_{m\in\mathbb{Z}}|T(e_k,e_m)|^2)^{1/2}\in Y$ and

$$\|y\|_Y\leqslant A_p^{-2}M^{(p)}(Y)\|T\|_{c_0\times c_0\to Y}.$$

Since $(\sum_{m\in\mathbb{Z}}\sum_{k\in\mathbb{Z}}|T(e_k,e_{m-k})|^2)^{1/2}\in Y$, it follows that $(\sum_{k\in\mathbb{Z}}|T(e_k,e_{m-k})|^2)^{1/2}\in Y$ for each $m\in\mathbb{Z}$. Hence, by the Cauchy–Schwarz inequality, the series $\sum_{k\in\mathbb{Z}}a_{km}T(e_k,e_{m-k})$ converges in Y for each $m\in\mathbb{Z}$. Now suppose $\{I_n\}_{n\in\mathbb{Z}}$ is any sequence of pairwise disjoint sets $I_n\subset\mathbb{Z}$ and $\{\delta_m\}_{m\in\mathbb{Z}}$ is any sequence of real numbers with $|\delta_m|\leqslant 1$ for each $m\in\mathbb{Z}$. To finish the proof it is enough to observe that the Cauchy–Schwartz inequality in combination with our hypothesis on $\{a_{km}\}$ implies the pointwise estimates

$$\begin{aligned}\left(\sum_{n\in\mathbb{Z}}\left|\sum_{m\in I_n}\delta_m y_m\right|^2\right)^{1/2} &\leqslant \Big(\sum_{n\in\mathbb{Z}}\sum_{m\in I_n}\left(\sum_{k\in\mathbb{Z}}|a_{km}|^2\right)\left(\sum_{k\in\mathbb{Z}}\big|T(e_k,e_{m-k})\big|^2\right)^{1/2} \\ &\leqslant \sup_{m\in\mathbb{Z}}\big\|\{a_{km}\}_k\big\|_{\ell_2}\left(\sum_{n\in\mathbb{Z}}\sum_{m\in I_n}\sum_{k\in\mathbb{Z}}\big|T(e_k,e_{m-k})\big|^2\right)^{1/2} \\ &\leqslant C\left(\sum_{k\in\mathbb{Z}}\sum_{m\in\mathbb{Z}}\big|T(e_k,e_m)\big|^2\right)^{1/2},\end{aligned}$$

which completes the proof. □

3. Interpolation of bilinear operators

In this section we will prove the main results of the paper. To begin a discussion we need some definitions. We use the standard notion and definitions from the interpolation theory which can be found, e.g., in [4] and [7]. For a given quasi-Banach couple $\vec{A}=(A_0,A_1)$, we define quasi-Banach spaces $A_0\cap A_1$ and A_0+A_1 equipped with the natural norms. A quasi-Banach space A is called intermediate with respect to $\vec{A}$ provided $A_0\cap A_1\subset A\subset A_0+A_1$. Let $\vec{X}=(X_0,X_1)$, $\vec{Y}=(Y_0,Y_1)$ and $\vec{Z}=(Z_0,Z_1)$ be quasi-Banach couples. For every bilinear operator $T:=(T_0,T_1)$ from $\vec{X}\times\vec{Y}$ into $\vec{Z}$ a bilinear operator T^0 (resp., T^1), called a *natural bilinear extension* of (T_0,T_1) from $(X_0+X_1)\times(Y_0\cap Y_1)$ into Z_0+Z_1 (resp., $(X_0\cap X_1)\times(Y_0+Y_1)\to Z_0+Z_1$) by

$$T^0(x,y):=T_0(x_0,y)+T_1(x_1,y)$$

for any $x=x_0+x_1$ and $y\in Y_0\cap Y_1$ (resp., $T^1(x,y):=T_0(x,y_0)+T_1(x,y_1)$ for any $x\in X_0\cap X_1$ and $y=y_0+y_1\in Y_0+Y_1$ with $y_0\in Y_0$ and $y_1\in Y_1$). It is easy to see that T^0 (resp., T^1) does not depend on the representations of $x\in X_0+X_1$ (resp., $y\in Y_0+Y_1$).

Throughout the paper $\mathcal{P}$ denotes the set of all positive quasi-concave functions ρ on $(0,\infty)$, i.e. such that both ρ and $t\mapsto t\rho(1/t)$ are non-decreasing functions. We let $\mathcal{P}_0$ denote the subset of $\mathcal{P}$ consisting of all ρ such that $\rho(t)\to 0$ as $t\to 0+$, and $\rho(t)/t\to\infty$ as $t\to\infty$. On $\mathcal{P}$, we define an involution by $\rho^*(t)=1/\rho(1/t)$ for every $t>0$ and we put $\mathcal{P}^*:=\mathcal{P}_0\cap(\mathcal{P}_0)^*$. We first prove an interpolation theorem for bilinear operators from corresponding couples of weighted c_0 spaces generated by quasi-concave functions into any Banach couple. We start with the following lemma.

Lemma 3.1. *Let $\rho_0,\rho_1,\rho\in\mathcal{P}_0$ be such that $\{\rho(2^n)\}\in\ell_1+\ell_1(2^{-n})$ and $\rho(st)\geqslant C\rho_0(s)\rho_1(t)$ for some $C>0$ and for every $s,t>0$. Assume that (Y_0,Y_1) is a Banach couple and*

$$T:\big(c_0,c_0\big(2^{-n}\big)\big)\times\big(c_0,c_0\big(2^{-n}\big)\big)\to(Y_0,Y_1)$$

is a bilinear operator. Then we have

$$\sum_{m\in\mathbb{Z}}\bigg\|\sum_{k\in\mathbb{Z}}\xi_k\eta_{m-k}T_0(e_k,e_{m-k})\bigg\|_{Y_0+Y_1}<\infty$$

for every sequences $\{\xi_n\}$ and $\{\eta_n\}$ in $c_0\cap c_0(2^{-n})$.

Proof. Fix $\xi = \{\xi_n\}$, $\eta = \{\eta_n\}$ in $c_0 \cap c_0(2^{-n})$ such that $\|\xi\|_{c_0(1/\rho_0(2^n))} \leqslant 1$, $\|\eta\|_{c_0(1/\rho_1(2^n))} \leqslant 1$. Let $C_j = \|T_j\|_{c_0(2^{-jn}) \times c_0(2^{-jn}) \to Y_j}$ for $j = 0, 1$. Our hypothesis implies $\rho_0(2^k)\rho_1(2^{m-k}) \leqslant C\rho(2^m)$ for all $k, m \in \mathbb{Z}$. Hence, it follows by Lemma 2.1(i),

$$\left\| \sum_{k\in\mathbb{Z}} \xi_k \eta_{m-k} T_0(e_k, e_{m-k}) \right\|_{Y_0} \leqslant C_0 \sup_{k\in\mathbb{Z}} |\xi_k \eta_{m-k}| \leqslant CC_0 \rho\big(2^m\big).$$

We set $S(s,t) := T_1(\{2^n s_n\}, \{2^n t_n\})$ for every $s = \{s_n\} \in c_0$, $t = \{t_n\} \in c_0$. Then S is a bilinear operator from $c_0 \times c_0$ into Y_1 with $\|S\|_{c_0 \times c_0 \to Y_1} \leqslant C_1$. Combining with similar estimates for S, we obtain

$$\begin{aligned} \left\| \sum_{k\in\mathbb{Z}} \xi_k \eta_{m-k} T_1(e_k, e_{m-k}) \right\|_{Y_1} &= 2^{-m} \left\| \sum_{k\in\mathbb{Z}} \xi_k \eta_{m-k} S(e_k, e_{m-k}) \right\|_{Y_1} \\ &\leqslant C_1 2^{-m} \sup_{k\in\mathbb{Z}} |\xi_k \eta_{m-k}| \leqslant CC_1 \frac{\rho(2^m)}{2^m}. \end{aligned}$$

Since $T_0(e_i, e_j) = T_1(e_i, e_j)$ for all $i, j \in \mathbb{Z}$, the above estimates yield

$$\begin{aligned} &\sum_{m\in\mathbb{Z}} \left\| \sum_{k\in\mathbb{Z}} \xi_k \eta_{m-k} T_0(e_k, e_{m-k}) \right\|_{Y_0+Y_1} \\ &\quad \leqslant \sum_{m<0} \left\| \sum_{k\in\mathbb{Z}} \xi_k \eta_{m-k} T_0(e_k, e_{m-k}) \right\|_{Y_0} + \sum_{m\geqslant 0} \left\| \sum_{k\in\mathbb{Z}} \xi_k \eta_{m-k} T_1(e_k, e_{m-k}) \right\|_{Y_1} \\ &\quad \leqslant C \max\{C_0, C_1\} \left(\sum_{m<0} \rho\big(2^m\big) + \sum_{m\geqslant 0} 2^{-m} \rho\big(2^m\big) \right) \\ &\quad = C \max\{C_0, C_1\} \big\| \big\{\rho\big(2^m\big)\big\} \big\|_{\ell_1 + \ell_1(2^{-m})} \end{aligned}$$

and the proof is complete. $\square$

We recall that if X is a quasi-Banach space, then X is said to be p-normable if there exists a constant C such that for any $x_1, \ldots, x_n \in X$ we have

$$\left\| \sum_{j=1}^n x_j \right\|_X \leqslant C \left(\sum_{j=1}^n \|x_j\|_X^p \right)^{1/p}.$$

It is well known that a fundamental theorem of Aoki and Rolewicz (see [22, Theorem 1.3]) asserts that every quasi-Banach space is p-normable for some $0 < p \leqslant 1$.

Further, we will use the following lemma (see [17, Lemma 3.1]).

Lemma 3.2. *Let X be a p-normable Banach space and let $\{x_{k,m}\}$ be an infinite matrix in X with $k, m \in \mathbb{Z}$. Assume that the series $\sum_{k\in\mathbb{Z}} x_{k,m-k}$ is unconditionally convergent for every $m \in \mathbb{Z}$ and $\sum_{m\in\mathbb{Z}} \| \sum_{k\in\mathbb{Z}} x_{k,m-k}\|_X^p < \infty$. Then the double limit* $\lim_{M,N\to\infty} \sum_{|k|\leqslant M} \sum_{|j|\leqslant N} x_{k,j}$ *exists in X and*

$$\lim_{M,N\to\infty}\sum_{|k|\leqslant M}\sum_{|j|\leqslant N}x_{k,j}=\sum_{m\in\mathbb{Z}}\biggl(\sum_{k\in\mathbb{Z}}x_{k,m-k}\biggr).$$

We will need the definition of an interpolation method which will appear in a natural way in our further results. For every Banach couple (X_0, X_1) and $\rho \in \mathcal{P}_0$ with $\{\rho(2^n)\}_{n\in\mathbb{Z}} \in \ell_2 + \ell_2(2^{-n})$ the space $G_{\rho,2}(X_0, X_1)$ consists of all elements $x \in X_0 + X_1$ for which $x = \sum_{n\in\mathbb{Z}} x_n$ (convergence in $X_0 + X_1$), where the elements $x_n \in X_0 \cap X_1$ are such that $\{2^{jn}x_n/\rho(2^n)\} \in \ell_2^w(X_j)$ for $j = 0, 1$. It is easy to verify that $G_{\rho,2}(X_0, X_1)$ is a Banach space equipped with the norm

$$\|x\| = \inf \max_{j=0,1}\bigl\|\bigl\{2^{jn}x_n/\rho\bigl(2^n\bigr)\bigr\}\bigr\|_{\ell_2^w(X_j)},$$

where the infimum is taken over all representations of $x = \sum_{n\in\mathbb{Z}} x_n$ as above.

We omit the simple proof that $G_{\rho,2}$ is an interpolation method. Notice that Proposition 3.1 stated below gives the so called an orbital description of $G_{\rho,2}(X_0, X_1)$ and so interpolation property follows.

We are now ready for our first bilinear interpolation theorem.

Theorem 3.1. *Let $\rho_0, \rho_1, \rho \in \mathcal{P}_0$ be such that $\{\rho(2^n)\} \in \ell_1 + \ell_1(2^{-n})$ and $\rho(st) \geqslant C\rho_0(s)\rho_1(t)$ for some $C > 0$ and all $s, t > 0$. Assume that (Y_0, Y_1) is a Banach couple and $T : (c_0, c_0(2^{-n})) \times (c_0, c_0(2^{-n})) \to (Y_0, Y_1)$ is a bilinear operator. Then T extends to a bilinear operator*

$$\widetilde{T} : c_0\bigl(1/\rho_0\bigl(2^n\bigr)\bigr) \times c_0\bigl(1/\rho_1\bigl(2^n\bigr)\bigr) \to G_{\rho,2}(Y_0, Y_1)$$

with $\|\widetilde{T}\| \leqslant \kappa_L A_1^{-1} C \max_{j=0,1} \|T_j\|_{c_0(2^{-jn})\times c_0(2^{-jn})\to Y_j}$.

Proof. Let $T := (T_0, T_1)$. Fix $\xi = \{\xi_n\}$, $\eta = \{\eta_n\} \in c_0 \cap c_0(2^{-n})$ with $\|\xi\|_{c_0(1/\rho_0(2^n))} \leqslant 1$ and $\|\eta\|_{c_0(1/\rho_1(2^n))} \leqslant 1$. Our hypothesis implies that the double positive sequence $\{a_{km}\}_{k,m\in\mathbb{Z}}$ defined by

$$a_{km} = \frac{\rho_0(2^k)\rho_1(2^{m-k})}{\rho(2^m)}, \quad k, m \in \mathbb{Z}$$

satisfies $\sup_{k,m\in\mathbb{Z}} a_{km} \leqslant C$. Since $T_0 : c_0 \times c_0 \to Y_0$ is a bilinear operator and $\|\{\xi_n/\rho_0(2^n)\}\|_{c_0} \leqslant 1$, $\|\{\xi_n/\rho_1(2^n)\}\|_{c_0} \leqslant 1$ and

$$u_m := \frac{1}{\rho(2^m)}\sum_{k\in\mathbb{Z}}\xi_k\eta_{m-k}T_0(e_k, e_{m-k}) = \sum_{k\in\mathbb{Z}} a_{km}\frac{\xi_k}{\rho(2^k)}\frac{\eta_{m-k}}{\rho_1(2^{m-k})}T_0(e_k, e_{m-k}),$$

we conclude by Lemma 2.1(ii) that $\{u_m\}_{m\in\mathbb{Z}}$ is weakly 2-summing sequence in Y_0 with

$$\bigl\|\{u_m\}\bigr\|_{\ell_2^w(Y_0)} \leqslant \kappa_L C\|T\|_{c_0\times c_0\to Y_0}.$$

Now, if $S(s,t) := T_1(\{2^n t_n\}, \{2^n s_n\})$ for $t = \{t_n\} \in c_0$ and $s = \{s_n\} \in c_0$, then $S : c_0 \times c_0 \to Y_1$ is a bilinear operator with $\|S\|_{c_0\times c_0\to Y_1} \leqslant \|T_1\|_{c_0(2^{-n})\times c_0(2^{-n})\to Y_1}$. Similar arguments to those shown above imply that, if for each $m \in \mathbb{Z}$, we set

$$v_m := \frac{2^m}{\rho(2^m)} \sum_{k\in\mathbb{Z}} \xi_k \eta_{m-k} T_1(e_k, e_{m-k}) = \sum_{k\in\mathbb{Z}} a_{km} \frac{\xi_k}{\rho(2^k)} \frac{\eta_{m-k}}{\rho_1(2^{m-k})} T_1(e_k, e_{m-k}),$$

then $\{v_m\}_{m\in\mathbb{Z}}$ is weakly 2-summing sequence in Y_1 and

$$\big\|\{v_m\}\big\|_{\ell_2^w(X_1)} \leqslant \kappa_L C \|T_1\|_{c_0(2^{-n})\times c_0(2^{-n})\to Y_1}.$$

Since (T_0, T_1) has a natural bilinear extension, it can be easily seen that

$$T_0(\xi, \eta) = \lim_{M,N\to\infty} \sum_{|k|\leqslant M} \sum_{|j|\leqslant N} \xi_k \eta_j T_0(e_k, e_j),$$

where the limit exists in $Y_0 + Y_1$. Now applying Lemmas 3.1 and 3.2, we conclude that

$$T_0(\xi, \eta) = \sum_{m\in\mathbb{Z}} \bigg(\sum_{k\in\mathbb{Z}} \xi_k \eta_{m-k} T_0(e_k, e_{m-k}) \bigg),$$

where the series converges in $Y_0 + Y_1$. Now, combining the above estimates, we obtain that $T_0(\xi, \eta) \in G_{\rho,2}(Y_0, Y_1)$ with

$$\begin{aligned}\big\|T_0(\xi, \eta)\big\|_{G_{\rho,2}(\vec{Y})} &\leqslant \max\big\{\big\|\{u_m\}\big\|_{\ell_2^w(Y_0)}, \big\|\{v_m\}\big\|_{\ell_2^w(Y_1)}\big\} \\ &\leqslant \kappa_L C \max_{j=0,1} \|T_j\|_{c_0(2^{-jn})\times c_0(2^{-jn})\to Y_j}.\end{aligned}$$

Since $c_0 \cap c_0(2^{-n})$ is dense in $c_0(1/\rho_j(2^n))$ for $j = 0, 1$, we are done. $\square$

We now present an extension of the above result to Peetre's interpolation method $\langle\cdot\rangle_\rho$. We recall that for each Banach couple (X_0, X_1) and each $\rho \in \mathcal{P}_0$, $\langle X_0, X_1\rangle_\rho$ is the space of all elements $x \in X_0 + X_1$ such that $x = \sum_{n\in\mathbb{Z}} x_n$ (convergence in $X_0 + X_1$), where the elements $x_n \in X_0 \cap X_1$ are such that $\sum_{n\in\mathbb{Z}} x_n/\rho(2^n)$ is unconditionally convergent in X_0 and $\sum_{n\in\mathbb{Z}} 2^n x_n/\rho(2^n)$ is unconditionally convergent in X_1. $\langle X_0, X_1\rangle_\rho$ is equipped with the norm

$$\|x\| = \inf \max_{j=0,1} \sup \bigg\| \sum_{n\in\mathbb{Z}} \lambda_n 2^{jn} x_n/\rho\big(2^n\big) \bigg\|_{X_j},$$

where the supremum is taken over all complex valued sequences $\{\lambda_n\}$ with $|\lambda_n| \leqslant 1$ for all n, and the infimum is taken over all representations of $x = \sum_{n\in\mathbb{Z}} x_n$. It is known that $\langle\cdot\rangle_\rho$ is equivalent to the minimal interpolation method G with the property $G(c_0, c_0(2^{-n})) \supset c_0(1/\rho(2^n))$ (see [20, Theorem 5]). Combining this fact with [29, Theorem 3.1], we immediately obtain the following result:

Theorem 3.2. *Let $\rho_0, \rho_1 \in \mathcal{P}^*$ and $\rho \in \mathcal{P}_0$ be such that $\{\rho(2^n)\} \in \ell_1 + \ell_1(2^{-n})$ and $\rho(st) \geqslant C\rho_0(s)\rho_1(t)$ for some $C > 0$ and for every $s, t > 0$. Assume that $T : (X_0, X_1) \times (Y_0, Y_1) \to (Z_0, Z_1)$ is a bilinear operator between Banach couples. Then T extends to a bilinear operator*

$$\widetilde{T} : \langle X_0, X_1\rangle_{\rho_0} \times \langle Y_0, Y_1\rangle_{\rho_1} \to G_{\rho,2}(Z_0, Z_1).$$

We wish to show some applications of the obtained results. We first note that in some cases spaces $G_{\rho,2}(X_0, X_1)$ have a natural description which is very useful for applications. We recall, following Aronszajn and Gagliardo [1], that given a Banach couple $\vec{A} = (A_0, A_1)$, $0 \neq a \in A_0 + A_1$ and any Banach couple $\vec{X}$ an *interpolation orbit* of a in $\vec{X}$ is a Banach space

$$\mathrm{Orb}_{\vec{A}}(a, \vec{X}) := \{Ta;\ T : \vec{A} \to \vec{X}\}$$

equipped with the norm

$$\|x\| = \inf\bigl\{\|T\|_{\vec{A}\to\vec{X}};\ x = Ta,\ T : \vec{A} \to \vec{X}\bigr\}.$$

It is well known that $\mathrm{Orb}_{\vec{A}}(a, \cdot)$ is an exact interpolation functor (see [1,7]).

The following proposition is an immediate consequence of an obvious well-known observation that the correspondence $T \mapsto \{Te_n\}_{n\in\mathbb{Z}}$ is an isometric isomorphism of the Banach space $L(\ell_2, X)$ onto $\ell_2^w(X)$ (see [12, Proposition 2.2]).

Proposition 3.1. *Let ρ be a quasi-concave function such that $\xi_\rho := \{\rho(2^n)\}_{n\in\mathbb{Z}} \in \ell_2 + \ell_2(2^{-n})$. Then for any Banach couple $\vec{X}$, we have*

$$G_{\rho,2}(\vec{X}) = \mathrm{Orb}_{(\ell_2,\ell_2(2^{-n}))}(\xi_\rho, \vec{X})$$

isometrically.

We now show how our results apply to Calderón–Lozanovskii spaces. Let Φ be the set of all functions $\varphi : \mathbb{R}_+ \times \mathbb{R}_+ \to \mathbb{R}_+$ that are positive, non-decreasing in each variable, and homogeneous of degree one (that is, $\varphi(\lambda s, \lambda t) = \lambda\varphi(s,t)$ for all $\lambda, s, t \geqslant 0$). Let $\varphi \in \Phi$ and $\vec{X} = (X_0, X_1)$ be a couple of quasi-Banach lattices on a measure space (Ω, μ). Following Lozanovskii [28], we define the space $\varphi(\vec{X}) = \varphi(X_0, X_1)$ of all $x \in L^0(\mu)$ such that $|x| = \varphi(|x_0|, |x_1|)$ for some $x_j \in X_j$, $j = 0, 1$. It is a quasi-Banach lattice equipped with the quasi-norm

$$\|x\| = \inf\bigl\{\max\bigl\{\|x_0\|_{X_0}, \|x_1\|_{X_1}\bigr\};\ |x| = \varphi(x_0, x_1), x_j,\ j = 0, 1\bigr\}.$$

Note that if φ is concave and (X_0, X_1) is a Banach couple, then $\varphi(\vec{X})$ is a Banach lattice which was studied by Lozanovskii [28] (see also [35] and references given therein).

A quasi-concave function ρ is called a *quasi-power* ($\rho \in \mathcal{P}^{+-}$) if $s_\rho(t) = o(\max(1,t))$ as $t \to 0$ and $t \to \infty$, where $s_\rho(t) := \sup_{u>0}(\rho(tu)/\rho(u))$ for every $t > 0$. The set of all $\varphi \in \Phi$ such that $\rho := \varphi(1, \cdot) \in \mathcal{P}_0$ (resp., $\rho \in \mathcal{P}^{+-}$, $\rho \in \mathcal{P}^*$) is denoted by Φ_0 (resp., Φ^{+-}, Φ^*). Throughout the paper the symbol $\varphi_1 \sim \varphi_2$ ($\varphi_1 \overset{0}{\sim} \varphi_2$ or $\varphi_1 \overset{\infty}{\sim} \varphi_2$) will mean that there exist C_1, C_2 and t_0 or t_∞ such that $C_1\varphi_1(t) \leqslant \varphi_2(t) \leqslant C_2\varphi_1(t)$ for all $t > 0$ ($0 < t \leqslant t_0$ or $t \geqslant t_\infty$, respectively).

Theorem 3.3. *Let $\varphi \in \Phi^{+-}$ be a concave function and let $\rho(t) = \varphi(1,t)$ for all $t > 0$. Then for any couple $(L_{p_0}(w_0), L_{p_1}(w_1))$ of weighted L_p-spaces on a measure space, we have*

$$G_{\rho,2}\bigl(L_{p_0}(w_0), L_{p_1}(w_1)\bigr) = \varphi\bigl(L_{r_0}(w_0), L_{r_1}(w_1)\bigr),$$

where $1/r_j = \max\{0, 1/p_j - 1/2\}$ for $j = 0, 1$.

Proof. Since $\varphi \in \Phi^{+-}$, it is easy to verify that for $\xi_\rho := \{\rho(2^n)\}$ the following equivalence holds:

$$K\bigl(t, \xi_\rho; \ell_2, \ell_2\bigl(2^{-n}\bigr)\bigr) \sim \rho(t).$$

Thus, we can apply the description of interpolation orbits in couples of the weighted L_p-spaces from [34] (see also [30]), which yields

$$\mathrm{Orb}_{(\ell_2, \ell_2(2^{-n}))}\bigl(\xi_\rho, \bigl(L_{p_0}(w_0), L_{p_1}(w_1)\bigr)\bigr) = \varphi\bigl(L_{r_0}(w_0), L_{r_1}(w_1)\bigr),$$

where $1/r_j = \max\{0, 1/p_j - 1/2\}$ for $j = 0, 1$. Since $\varphi \in \Phi^{+-}$, $\{\xi_\rho\} \in \ell_1 + \ell_1(2^{-n})$. Thus we may apply Proposition 3.3 to conclude the proof. $\square$

It is well known (see [31, Theorem 2.1]) that for any couple (X_0, X_1) of Banach lattices and any concave function $\varphi \in \Phi_0$ such that $\varphi^* \in \Phi_0$, we have

$$\langle X_0, X_1\rangle_\rho = \varphi(X_0, X_1)^\circ,$$

where the right hand of equality denotes the closure of $X_0 \cap X_1$ in $\varphi(X_0, X_1)$. If $\varphi \in \Phi^{+-}$ and X_j has an order continuous norm for $j = 0$ or $j = 1$, then $\varphi(X_0, X_1)$ has also an order continuous norm (see, e.g., [35]). In particular this implies that $X_0 \cap X_1$ is dense in $\varphi(X_0, X_1)$, i.e.,

$$\varphi(X_0, X_1)^\circ = \varphi(X_0, X_1).$$

Combining these remarks with the above results, we obtain a bilinear interpolation theorem for Calderón–Lozanovskii spaces. For simplicity, we present a special case.

Theorem 3.4. *Let $\varphi \in \Phi^{+-}$ be a concave function such that $\varphi(1, s)\varphi(1, t) \leqslant C\varphi(1, st)$ for some $C > 0$ and every $s, t > 0$. If $T : (X_0, X_1) \times (Y_0, Y_1) \to (L_{p_0}(w_0), L_{p_1}(w_1))$ is a bilinear operator, then T extends to a bilinear operator*

$$\widetilde{T} : \varphi(X_0, X_1)^\circ \times \varphi(Y_0, Y_1)^\circ \to \varphi\bigl(L_{r_0}(w_0), L_{r_1}(w_1)\bigr),$$

where $1/r_j = \max\{0, 1/p_j - 1/2\}$ for $j = 0, 1$.

Applying the above theorem to L_p-spaces we obtain bilinear interpolation theorem for Orlicz spaces. We recall that if ψ is an Orlicz function (i.e., $\psi : [0, \infty) \to [0, \infty)$ is an increasing, continuous such that $\psi(0) = 0$), then the Orlicz space L_ψ on a given measure space (Ω, μ) is defined to be a subspace of $L^0(\mu)$ consisting of all $f \in L^0(\mu)$ such that for some $\lambda > 0$ holds

$$\int_\Omega \psi\bigl(\lambda|f|\bigr)\, d\mu < \infty.$$

It is easy to check that if there exists $C > 0$ such that $\psi(t/C) \leqslant \psi(t)/2$ for all $t > 0$, then L_ψ is a quasi-Banach lattice with the quasi-norm $\|\cdot\|$ satisfying

$$\|f + g\| \leqslant C\bigl(\|f\| + \|g\|\bigr), \quad f, g \in L_\psi,$$

where

$$\|f\| := \inf\left\{\lambda > 0;\ \int_{\Omega} \psi\big(|f|/\lambda\big)\, d\mu \leqslant 1\right\}.$$

It is well known (see [31], [33, pp. 460–461]) that for any function $\varphi \in \Phi$ and any couple $(L_{p_0}(w_0), L_{p_1}(w_1))$ on a measure space (Ω, μ) with $0 < p_0, p_1 \leqslant \infty$

$$\varphi\big(L_{p_0}(w_0), L_{p_1}(w_1)\big) = L_{\mathcal{M}}$$

with equivalence of the quasi-norms, where $L_{\mathcal{M}}$ is the generalized Orlicz space generated by the function $\mathcal{M}(u,t) = \psi((w_1(t)^{1/p_1} w_0(t)^{-1/p_0})^q)(w_0(t)/w_1(t))^q$ for all $u \geqslant 0$ and $t \in \Omega$. Here $1/q = 1/p_0 - 1/p_1$ and ψ is an Orlicz function given by $\psi^{-1}(t) = \varphi(t^{1/p_0}, t^{1/p_1})$ for all $t > 0$. $L_{\mathcal{M}}$ is equipped with the quasi-norm

$$\|f\|_{\mathcal{M}} = \inf\left\{\lambda > 0;\ \int_{\Omega} \mathcal{M}\big(\big|f(t)\big|/\lambda, t\big)\, d\mu \leqslant 1\right\}.$$

In particular we have

$$\varphi(L_{p_0}, L_{p_1}) = L_{\psi}.$$

A simple calculation shows (see [33, p. 459]) that in the case $0 < p_0 = p_1 = p \leqslant \infty$ we have

$$\varphi\big(L_p(w_0), L_p(w_1)\big) = L_p\big(1/\varphi(w_0, w_1)\big).$$

Combining our bilinear interpolation theorems with shown above formulas, we obtain bilinear theorems for generalized Orlicz spaces. In particular as an application of Theorems 3.1 and 3.4 we state a special corollary that is interesting on its own.

Corollary 3.1. *Let $\rho_0, \rho_1, \rho \in \mathcal{P}_0$ be such that $\{\rho(2^n)\} \in \ell_1 + \ell_1(2^{-n})$ and $\rho(st) \geqslant C\rho_0(s)\rho_1(t)$ for some $C > 0$ and every $s, t > 0$. If $T : (c_0, c_0(2^{-n})) \times (c_0, c_0(2^{-n})) \to (\ell_1, \ell_1(2^n))$, then T extends to a bilinear operator*

$$\widetilde{T} : c_0\big(1/\rho_0\big(2^n\big)\big) \times c_0\big(1/\rho_1\big(2^n\big)\big) \to \ell_2\big(1/\rho\big(2^{-n}\big)\big).$$

Corollary 3.2. *Let $\varphi \in \Phi^{+-}$ be such that $\varphi(1,s)\varphi(1,t) \leqslant C\varphi(1,st)$ for some $C > 0$ and every $s, t > 0$. If $T : (c_0, c_0(2^{-n})) \times (c_0, c_0(2^{-n})) \to (\ell_1, \ell_1(2^n))$, then T extends to a bilinear operator*

$$\widetilde{T} : c_0\big(1/\rho\big(2^n\big)\big) \times c_0\big(1/\rho\big(2^n\big)\big) \to \ell_2\big(1/\rho\big(2^{-n}\big)\big),$$

where $\rho(t) = \varphi(1,t)$ for all $t > 0$.

Proof. It is easy to check that $\varphi \in \Phi^{+-}$ is equivalent to

$$\sum_{n \in \mathbb{Z}} \min\{1, 2^{-n}\} s_\rho(2^n) < \infty.$$

This implies that $\{\rho(2^n)\}_{n\in\mathbb{Z}} \in \ell_1 + \ell_1(2^{-n})$ and so Corollary 3.2 applies. $\square$

In the sequel we will use the following lemma (see [31, Lemma 2.2]), which is a general variant of Carlson's inequality proved in [19, Proposition 3.1] for every $\varphi \in \Phi^{+-}$.

Lemma 3.3. *Let $\varphi \in \Phi^*$. There exist a constant $C = C(\varphi)$ a sequence $\{I_n\}_{n\in\mathbb{Z}}$ of disjoint sets $I_n \subset \mathbb{Z}$ and two sequences $\{\delta_n^j\}_{n\in\mathbb{Z}}$ of real numbers with $0 \leqslant \delta_n^j \leqslant 1$ for each $n \in \mathbb{Z}$, $j = 0, 1$, such that*

$$\left|\sum_m a_m \varphi(1, 2^m)\right| \leqslant C\varphi\left(\left(\sum_n \left|\sum_{m\in I_n} \delta_m^0 a_m\right|^2\right)^{1/2}, \left(\sum_n \left|\sum_{m\in I_n} \delta_m^1 2^m a_m\right|^2\right)^{1/2}\right).$$

We note that when $\varphi \in \Phi^{+-}$, one may take $I_n = \{n\}$ for every $n \in \mathbb{Z}$ (see [19, p. 38]).

Theorem 3.5. *Let $\varphi_0, \varphi_1, \varphi \in \mathcal{P}^*$ and $\varphi \in \Phi^*$ be such that $\{\varphi(1, 2^n)\} \in \ell_p + \ell_p(2^{-n})$ for some $0 < p \leqslant 1$ and let $\sup_{m\in\mathbb{Z}}(\sum_{k\in\mathbb{Z}}(\varphi_0(1, 2^k)\varphi_1(1, 2^{m-k})/\varphi(1, 2^m))^2)^{1/2} < \infty$. Assume that $T : (X_0, X_1) \times (Y_0, Y_1) \to (Z_0, Z_1)$ is a bilinear operator between couples of quasi-Banach lattices. If both Z_0 and Z_1 are maximal p-normable with nontrivial convexity, then T extends to a bilinear operator*

$$\widetilde{T} : \varphi_0(X_0, X_1)^\circ \times \varphi_1(Y_0, Y_1)^\circ \to \varphi(Z_0, Z_1).$$

Proof. It is easy to adopt the proof of Lemma 8.2.1 in [32] to show that if $\psi \in \Phi_0$ and (E_0, E_1) is a couple of quasi-Banach lattices, then for any $x \in \psi(E_0, E_1)$ there exists an operator $S : (\ell_\infty, \ell_\infty(2^{-n})) \to (E_0, E_1)$ such that $x = Sa_\psi$, where $a_\psi = \{\psi(1, 2^n)\}$. This fact immediately implies that to prove our result it is enough to show that if $T : (c_0, c_0(2^{-n})) \times (c_0, c_0(2^{-n})) \to (Z_0, Z_1)$ is a bilinear operator, then it extends to a bilinear operator

$$\widetilde{T} : c_0\big(1/\rho_0(2^n)\big) \times c_0\big(1/\rho_1(2^n)\big) \to \varphi(Z_0, Z_1),$$

where $\rho_j = \varphi_j(1, \cdot)$ for $j = 0, 1$.

Let $T = (T_0, T_1)$. Take any $\xi = \{\xi_n\}$, $\eta = \{\eta_n\} \in c_0 \cap c_0(2^{-n})$ with $\|\xi\|_{c_0(1/\rho_0(2^n))} \leqslant 1$ and $\|\eta\|_{c_0(1/\rho_1(2^n))} \leqslant 1$. Our hypothesis implies that the double positive sequence $\{a_{km}\}_{k,m\in\mathbb{Z}}$ defined by

$$a_{km} = \frac{\xi_k \eta_{m-k}}{\rho(2^m)}, \quad k, m \in \mathbb{Z}$$

satisfies

$$C := \sup_{m\in\mathbb{Z}} \left(\sum_{k\in\mathbb{Z}} |a_{km}|^2\right)^{1/2} \leqslant \sup_{m\in\mathbb{Z}} \frac{1}{\rho(2^m)} \left(\sum_{k\in\mathbb{Z}} \big(\rho_0(2^k)\rho_1(2^{m-k})\big)^2\right)^{1/2} < \infty.$$

Since $T_0 : c_0 \times c_0 \to Z_0$ is a bilinear operator, we obtain by Lemma 2.2 that for each $m \in \mathbb{Z}$

$$u_m := \frac{1}{\rho(2^m)} \sum_{k\in\mathbb{Z}} \xi_k \eta_{m-k} T_0(e_k, e_{m-k}) = \sum_{k\in\mathbb{Z}} a_{km} T_0(e_k, e_{m-k}) \in Z_0$$

and similarly, by the fact that $T_1 : c_0(2^{-n}) \times c_0(2^{-n}) \to Z_1$ is a bilinear operator,

$$v_m := \frac{2^m}{\rho(2^m)} \sum_{k\in\mathbb{Z}} \xi_k \eta_{m-k} T_1(e_k, e_{m-k}) \in Z_1.$$

Now choose sequences $\{I_n\}_{n\in\mathbb{Z}}$ and $\{\delta_n^j\}_{n\in\mathbb{Z}}$, $j = 0, 1$, as in Lemma 3.3. Then again by Lemma 2.2, it follows that there exist constants C_0 and C_1 such that

$$\left\| \left(\sum_{n\in\mathbb{Z}} \left| \sum_{m\in I_n} \delta_m^0 u_m \right|^2 \right)^{1/2} \right\|_{Z_0} \leqslant C_0 \|T\|_{c_0\times c_0 \to Z_0} \tag{$*$}$$

and

$$\left\| \left(\sum_{n\in\mathbb{Z}} \left| \sum_{m\in I_n} \delta_m^1 v_m \right|^2 \right)^{1/2} \right\|_{Z_1} \leqslant C_1 \|T_1\|_{c_0(2^{-n})\times c_0(2^{-n}) \to Z_1}. \tag{$**$}$$

Our hypothesis yields $\rho(st) \geqslant C\rho_0(s)\rho_1(t)$, thus the proof of Lemma 3.1 shows that for some $C_1 > 0$ we have

$$\sum_{m\in\mathbb{Z}} \left\| \sum_{k\in\mathbb{Z}} \xi_k \eta_{m-k} T_0(e_k, e_{m-k}) \right\|_{Z_0+Z_1}^p \leqslant C_1 \left\| \left\{ \rho\left(2^m\right) \right\} \right\|_{\ell_p+\ell_p(2^{-m})}.$$

Since (T_0, T_1) has a natural bilinear extension, it can be explain essentially the same way (by Lemmas 3.1 and 3.2) as in the proof of Lemma 3.1 that

$$T_0(\xi, \eta) = \sum_{m\in\mathbb{Z}} \left(\sum_{k\in\mathbb{Z}} \xi_k \eta_{m-k} T_0(e_k, e_{m-k}) \right),$$

where the series converges in $Z_0 + Z_1$. Now Carlson's inequality implies

$$\left| T_0(\xi, \eta) \right| \leqslant C(\varphi)\varphi\left(\left(\sum_{n\in\mathbb{Z}} \left| \sum_{m\in I_n} \delta_m^0 u_m \right|^2 \right)^{1/2}, \left(\sum_{n\in\mathbb{Z}} \left| \sum_{m\in I_n} \delta_m^1 v_m \right|^2 \right)^{1/2} \right), \quad \text{a.e.}$$

and so $T_0(\xi, \eta) \in \varphi(Z_0, Z_1)$ by $(*)$ and $(**)$. The proof is complete. $\square$

For any $1 \leqslant p < \infty$ we define a subclass $\mathcal{P}_p$ of $\mathcal{P}$ containing of all $\rho \in \mathcal{P}$ such that

$$\sup_{m\in\mathbb{Z}} \frac{1}{\rho(2^m)} \left(\sum_{k\in\mathbb{Z}} \left(\rho\left(2^k\right) \rho\left(2^{m-k}\right) \right)^p \right)^{1/p} < \infty.$$

In the case when $\rho \in \mathcal{P}_2$, we obtain the following strengthening of Theorem 3.2.

Theorem 3.6. *Let $\varphi \in \Phi^{+-}$ be such that $\rho = \varphi(1, \cdot) \in \mathcal{P}_2$. Assume that $T : (X_0, X_1) \times (Y_0, Y_1) \to (Z_0, Z_1)$ is a bilinear operator between quasi-Banach lattices. If Z_0 and Z_1 have nontrivial convexity, then T extends to a bilinear operator*

$$\widetilde{T} : \varphi(X_0, X_1)^\circ \times \varphi(Y_0, Y_1)^\circ \to \varphi(Z_0, Z_1).$$

Proof. Since $\varphi \in \Phi^{+-}$, $\sum_{n\in\mathbb{Z}} \min\{1, 2^{-n}\} s_\rho(2^n) < \infty$. This implies that $\{\rho(2^n)\}_{n\in\mathbb{Z}} \in \ell_p + \ell_p(2^{-n})$ for every $0 < p \leqslant 1$. Thus the result follows by Theorem 3.5. □

We note that if in the statement of Corollary 3.2, we would have $\ell_1(1/\rho(2^{-n}))$ instead of $\ell_2(1/\rho(2^{-n}))$, then via [29, Theorem 3.1] the result could be lifted to more general case, where the range space would be generated by Ovchinnikov's upper method of interpolation. This fact motivates the question for which quasi-concave functions ρ the statement of Corollary 3.2 is true with $\ell_1(1/\rho(2^{-n}))$ instead of $\ell_2(1/\rho(2^{-n}))$? Below we present a class of quasi-concave functions for which the answer for this question is positive.

For a given $\varphi \in \Phi$, let φ_u denote the upper Ovchinnikov's functor (see [32]). This is the maximal interpolation functor satisfying $\varphi_u(\ell_1, \ell_1(2^n)) \subset \ell_1(1/\varphi(1, 2^{-n}))$, i.e., if $\vec{X} = (X_0, X_1)$ is any Banach couple, then the Banach space $\varphi_u(\vec{X})$ consists of all $x \in X_0 + X_1$ such that

$$\|x\|_{\varphi_u(\vec{X})} := \sup\Big\{\|Tx\|_{\ell_1(1/\varphi(1,2^{-n}))};\ \max_{j=0,1} \|T\|_{X_j \to \ell_1(2^{jn})} \leqslant 1\Big\}.$$

Theorem 3.7. *Let $\varphi \in \Phi^{+-}$ be such that $\rho = \varphi(1, \cdot) \in \mathcal{P}_2$. Assume that $T : (X_0, X_1) \times (Y_0, Y_1) \to (Z_0, Z_1)$ is a bilinear operator between Banach couples. Then T extends to a bilinear operator*

$$\widetilde{T} : \langle X_0, X_1\rangle_\rho \times \langle Y_0, Y_1\rangle_\rho \to \varphi_u(Z_0, Z_1).$$

Proof. From Theorem 3.6, it follows that every bilinear operator $T : (c_0, c_0(2^{-n})) \times (c_0, c_0(2^{-n})) \to (\ell_1, \ell_1(2^n))$ extends to a bilinear operator

$$\widetilde{T} : \varphi\big(c_0, c_0\big(2^{-n}\big)\big) \times \varphi\big(c_0, c_0\big(2^{-n}\big)\big) \to \varphi\big(\ell_1, \ell_1\big(2^n\big)\big),$$

i.e.,

$$\widetilde{T} : c_0\big(1/\rho\big(2^n\big)\big) \times c_0\big(1/\rho\big(2^n\big)\big) \to \ell_1\big(1/\rho\big(2^{-n}\big)\big).$$

Thus the results follows by [29, Theorem 3.1]. □

We give examples of functions satisfying a stronger condition than the one required in Theorem 3.6. Following [3], we define ϕ by

$$\phi(t) = \begin{cases} t^a \ln^c(C_1/t), & 0 < t \leqslant 1, \\ t^b \ln^d(C_2 t), & t > 1, \end{cases}$$

where $0 < a < b < 1$, $c > 1$, $d > 1$ and constants $C_1 > e^{c/a}$, $C_2 > e^d d/(1-b)$ are chosen such that ϕ is continuous. Then ϕ is a quasi-power and satisfies

$$\sup_{m\in\mathbb{Z}} \sum_{k\in\mathbb{Z}} \frac{\phi(2^m)}{\phi(2^k)\phi(2^{m-k})} < \infty.$$

In consequence ρ defined by $\rho(t) := t/\phi(t)$ for every $t > 0$ satisfies

$$\sup_{m\in\mathbb{Z}} \frac{1}{\rho(2^m)} \sum_{k\in\mathbb{Z}} \rho\big(2^k\big)\rho\big(2^{m-k}\big) < \infty,$$

and whence $\rho \in \mathcal{P}_2$.

The following Proposition shows that for two given functions in subclass $\mathcal{P}_p$ we can produce more general examples of functions from $\mathcal{P}_p$.

Proposition 3.2. *Suppose $\rho_0, \rho_1 \in \mathcal{P}_p$ for some $1 \leqslant p < \infty$. If $\varphi \in \Phi$ is such that $C\varphi(1, st) \geqslant \varphi(1, s)\varphi(1, t)$ for some $C > 0$ and all $s, t > 0$, then $\rho \in \mathcal{P}_p$, where $\rho(t) := \varphi(\rho_0(t), \rho_1(t))$ for all $t > 0$.*

Proof. If we define $\phi(s, t) := \varphi^p(s^{1/p}, t^{1/p})$ for every $s, t > 0$, then $\phi \in \Phi$ and so $\phi(s, t) \leqslant \max\{s/\alpha, t/\beta\}\phi(\alpha, \beta)$ for every $\alpha, \beta, s, t > 0$. This implies $\phi(s, t) \leqslant \widehat{\phi}(s, t) \leqslant 2\phi(s, t)$, where

$$\widehat{\phi}(s, t) = \inf_{\alpha,\beta>0} \max\left(\frac{s}{\alpha} + \frac{t}{\beta}\right)\phi(\alpha, \beta), \quad s, t > 0.$$

Thus, since $\widehat{\phi} \in \Phi$ is concave, we may assume without loss of the generality that ϕ is concave. Then we have

$$\sum_{j=1}^{n} \phi(s_j, t_j) \leqslant \phi\left(\sum_{j=1}^{n} s_j, \sum_{j=1}^{n} t_j\right)$$

for all positive numbers $s_1, \ldots, s_n$ and $t_1, \ldots, t_n$. Our hypothesis on φ implies that $C\varphi(uv, st) \geqslant \varphi(u, s)\varphi(v, t)$ for all $u, v, s, t > 0$. Combining these inequalities, it follows that if $\rho := \varphi(\rho_0, \rho_1)$, then for any finite subset F of $\mathbb{Z}$, we have

$$\begin{aligned}
\sum_{k\in F} \big(\rho\big(2^k\big)\rho\big(2^{m-k}\big)\big)^p &= \sum_{k\in F} \big(\varphi\big(\rho_0\big(2^k\big), \rho_1\big(2^k\big)\big)\varphi\big(\rho_0\big(2^{m-k}\big), \rho_1\big(2^{m-k}\big)\big)\big)^p \\
&\leqslant \sum_{k\in F} \big(C\varphi\big(\rho_0\big(2^k\big)\rho_0\big(2^{m-k}\big), \rho_1\big(2^k\big)\rho_1\big(2^{m-k}\big)\big)\big)^p \\
&= C^p \sum_{k\in F} \phi\big(\big(\rho_0\big(2^k\big)\rho_0\big(2^{m-k}\big)\big)^p, \big(\rho_1\big(2^k\big)\rho_1\big(2^{m-k}\big)\big)^p\big) \\
&\leqslant C^p \phi\left(\sum_{k\in F} \big(\rho_0\big(2^k\big)\rho_0\big(2^{m-k}\big)\big)^p, \sum_{k\in F} \big(\rho_1\big(2^k\big)\rho_1\big(2^{m-k}\big)\big)^p\right).
\end{aligned}$$

Thus if $\rho_0, \rho_1 \in \mathcal{P}_p$, then there exist $C_0, C_1 > 0$ such that for all $m \in \mathbb{Z}$, we have

$$\sum_{k\in\mathbb{Z}}\bigl(\rho(2^k)\rho(2^{m-k})\bigr)^p \leqslant C^p\phi\bigl((C_0\rho_0(2^m))^p, (C_1\rho_1(2^m))^p\bigr)$$
$$\leqslant C_2^p\varphi^p\bigl(\rho_0(2^m), \rho_1(2^m)\bigr) = \bigl(C_2\rho(2^m)\bigr)^p,$$

where $C_2 = C\max\{C_0, C_1\}$. This shows that $\rho \in \mathcal{P}_p$ and the proof is complete. □

4. Applications to the bilinear Hilbert transform

We now give some applications of our bilinear interpolation theorems. Motivated by applications to Zygmund classes, we use below a special variant of the bilinear interpolation theorem which is a consequence of Theorem 3.6 by the fact that any L_p-space is p-concave for every $0 < p < \infty$.

Theorem 4.1. *Assume that $\rho \in \mathcal{P}_2$, $1 \leqslant u_j, v_j < \infty$ and $0 < s_j < \infty$ for $j = 0, 1$ and let $T : (L_{u_0}, L_{u_1}) \times (L_{v_0}, L_{v_1}) \to (L_{s_0}, L_{s_1})$ be a bilinear operator between couples of L_p-spaces. Then T extends to a bounded bilinear operator*

$$\widetilde{T} : L_{\psi_0} \times L_{\psi_1} \to L_{\psi}$$

between Orlicz spaces with $\psi_0^{-1}(t) \sim t^{1/u_0}\rho(t^{1/u_1-1/u_0})$, $\psi_1^{-1}(t) \sim t^{1/v_0}\rho(t^{1/v_1-1/v_0})$ and $\psi^{-1}(t) \sim t^{1/s_0}\rho(t^{1/s_1-1/s_0})$.

We specialize the above theorem to the family of the bilinear Hilbert transforms $\{H_\theta\}_{\theta\in\mathbb{R}}$, however this can also be applied to bilinear multipliers described in the introduction. We recall that the bilinear Hilbert transform H_θ is defined for a parameter $\theta \in \mathbb{R}$ by

$$H_\theta(f, g)(x) := \lim_{\varepsilon\to 0}\int_{|t|>\varepsilon} f(x-t)g(x+\theta t)\frac{1}{t}\,dt, \quad x \in \mathbb{R}$$

for functions f, g from the Schwartz class $\mathcal{S}(\mathbb{R})$. The family $\{H_\theta\}$ was introduced by Calderón in his study of the first commutator, an operator arising in a series decomposition of the Cauchy integral along Lipschitz curves. In [9] Calderón posed the question whether H_θ satisfies any L_p estimates. In their fundamental work [25], Lacey and Thiele proved that if $\theta \neq -1$, then the bilinear Hilbert transform H_θ extends to a bilinear operator from $L_p \times L_q$ into L_r whenever $1 < p, q \leqslant \infty$ and $1/p + 1/q = 1/r < 3/2$. We also refer to a remarkable paper [16] of Grafakos and Li, where the uniform boundedness of the bilinear Hilbert transforms H_θ for a range of exponents is proved. It should be noted that the bilinear Hilbert transforms arise in a variety of other related known problems in bilinear Fourier analysis, e.g., in the study of the convergence of the mixed Fourier series of the form

$$\lim_{N\to\infty}\sum_{\substack{|m-\theta n|\leqslant N\\ |m-n|\leqslant N}}\sum \hat{f}(m)\hat{g}(n)e^{2\pi i(m+n)x}$$

for functions f, g defined on $[0, 2\pi]$. We also note that Fan and Sato [13] were able to show the boundedness of the bilinear Hilbert transform H on the torus $\mathbb{T}$

$$H(f,g)(x) := \int_{\mathbb{T}} f(x-t)g(x+t)\cot(\pi t)\,dt, \quad x \in \mathbb{T}$$

by transferring the result from $\mathbb{R}$. Their proof relies upon some DeLeeuw type transference methods for multilinear multipliers (see [11]).

Motivated by applications to Zygmund classes, we use below a special variant of the bilinear interpolation theorem which is a consequence of Theorem 3.6 by the fact that any L_p-space is p-convex for every $0 < p < \infty$.

An important class of Orlicz spaces are Zygmund classes. For $0 < \alpha, \beta < \infty$ and $0 < p, q < \infty$ the Zygmund space $\mathcal{Z}_{p,q}^{\alpha,\beta}$ on a measure space (Ω, μ) consists of all $f \in L^0(\mu)$ such that

$$\int_{\{|f| \leqslant 1\}} |f|^p \big(1 - \log|f|\big)^\alpha \, d\mu + \int_{\{|f|>1\}} |f|^q \big(1 + \log|f|\big)^\beta \, d\mu < \infty.$$

It is clear that if $\psi : [0,\infty) \to [0,\infty)$ is an Orlicz function such that $\psi(t) \overset{0}{\sim} t^p(1-\log t)^\alpha$ and $\psi(t) \overset{\infty}{\sim} t^q(1+\log t)^\beta$, then $\mathcal{Z}_{p,q}^{\alpha,\beta}$ coincides with L_ψ. In what follows we equipped $\mathcal{Z}_{p,q}^{\alpha,\beta}$ with the quasi-norm $\|\cdot\|_\psi$ generated by shown ψ. As an application of Theorem 4.1 and above mentioned results, we will show boundedness of the bilinear Hilbert transform between corresponding Zygmund classes.

Theorem 4.2. *For any $\theta \neq -1$ the bilinear Hilbert transform H_θ extends to a bilinear operator from $\mathcal{Z}_{p_0,q_0}^{\alpha_0,\beta_0} \times \mathcal{Z}_{p_1,q_1}^{\alpha_1,\beta_1}$ into $\mathcal{Z}_{p,q}^{\alpha,\beta}$ if $1 < p_0 < q_0 < \infty$, $1 < p_1 < q_1 < \infty$, $1/p_0 + 1/p_1 = 1/p < 3/2$, $\alpha_0/p_0 = \alpha_1/p_1 = \alpha/p > 1$ and $\beta_0/q_0 = \beta_1/q_1 = \beta/q > 1$.*

Proof. We have $c := \beta_0/q_0 = \beta_1/q_1 = \beta/q > 1$ and $d := \alpha_0/p_0 = \alpha_1/p_1 = \alpha/p > 1$. For $0 < b < a < 1$ define $\phi_{a,b} \in \mathcal{P}_1$ by

$$\phi_{a,b}(t) := \begin{cases} \frac{t^a}{\log^c(C_1/t)}, & 0 < t \leqslant 1, \\ \frac{t^b}{\log^d(C_2 t)}, & t > 1, \end{cases}$$

where C_1 and C_2 are chosen such that ρ is continuous.

Let ψ_j $(j = 0, 1)$ and ψ be Orlicz functions such that $\psi_j(t) \overset{0}{\sim} t^{p_j}(1-\log t)^{\alpha_j}$, $\psi_j(t) \overset{\infty}{\sim} t^{q_j}(1+\log t)^{\beta_j}$ and $\psi(t) \overset{0}{\sim} t^p(1-\log t)^\alpha$, $\psi(t) \overset{\infty}{\sim} t^q(1+\log t)^\beta$. Then, as it was explained, we have

$$L_{\psi_0} = \mathcal{Z}_{p_0,q_0}^{\alpha_0,\beta_0}, \qquad L_{\psi_1} = \mathcal{Z}_{p_1,q_1}^{\alpha_1,\beta_1}, \qquad L_\psi = \mathcal{Z}_{p,q}^{\alpha,\beta}.$$

An elementary computation shows that

$$\psi_j^{-1}(t) \overset{0}{\sim} \frac{t^{1/p_j}}{(1-\log t)^d}, \qquad \psi_j^{-1}(t) \overset{\infty}{\sim} t^{1/q_j}(1+\log t)^c, \quad j = 0, 1$$

and

$$\psi^{-1}(t) \overset{0}{\sim} \frac{t^{1/p}}{(1-\log t)^d}, \qquad \psi^{-1}(t) \overset{\infty}{\sim} t^{1/q}(1+\log t)^c.$$

We claim that our hypotheses imply that there exist $0 < b < a$, $1 < u_0 < u_1 < \infty$ and $1 < v_0 < v_1 < \infty$ such that the following equivalences hold with $\rho = \phi_{a,b}$

$$\psi_0^{-1}(t) \sim t^{1/u_0}\rho\big(t^{1/u_1-1/u_0}\big), \qquad \psi_1^{-1}(t) \sim t^{1/v_0}\rho\big(t^{1/v_1-1/v_0}\big)$$

and

$$\psi^{-1}(t) \sim t^{1/s_0}\rho\big(t^{1/s_1-1/s_0}\big),$$

where

$$1/s_1 = 1/u_1 + 1/v_1 < 1/s_0 = 1/u_0 + 1/v_0 < 3/2.$$

Before we prove the claim observe that combining the above statements with the mentioned result of Lacey and Thiele [25], we conclude that for any $\theta \neq -1$ the bilinear Hilbert transform H_θ extends to a bilinear operator

$$\widetilde{H}_\theta : (L_{u_0}, L_{u_1}) \times (L_{v_0}, L_{v_1}) \to (L_{s_0}, L_{s_1}).$$

Then the required result follows by Theorem 4.1.

All that remains is the proof of claim. In order to do this, we first observe that if $1 < p_0, p_1 < \infty$ and $1 < q_0, q_1 < \infty$, then for every $0 < b < a < 1$ the systems of equations for reals variables $u_0, u_1, v_0, v_1 \neq 0$

$$(1-a)/u_0 + a/u_1 = 1/p_0, \qquad (1-b)/u_0 + b/u_1 = 1/q_0,$$
$$(1-a)/v_0 + a/v_1 = 1/p_1, \qquad (1-b)/v_0 + b/v_1 = 1/q_1$$

have solutions. We choose a and b such that $a/b > \max\{p_0/q_0, p_0'/q_0', p_1/q_1, p_1'/q_1'\}$ (and resp., $(1-b)/(1-a) > \max\{q_0/p_0, q_0'/p_0', q_1/p_1, q_1'/p_1'\}$; here as usual if $1 < s < \infty$, the conjugate of s is the number which satisfies $1/s + 1/s' = 1$). Then it is easy to check that this choice of a and b and our hypotheses ensure that $1 < u_0 < u_1 < \infty$ by $p_0 < p_1$ (and resp., $1 < v_0 < v_1 < \infty$ by $p_1 < q_1$). This yields

$$1/s_1 := 1/u_1 + 1/v_1 < 1/s_0 := 1/u_0 + 1/v_0,$$

and whence

$$(1-b)/s_0 + b/s_1 = 1/p, \qquad (1-a)/s_0 + b/s_1 = 1/q.$$

If additionally a and b satisfy

$$\frac{a}{b} > \frac{3/2 - 1/q}{3/2 - 1/p},$$

then we get $2/3 < s_1 < s_0$.

It is evident that all shown above inequalities for a and b are true whenever a is chosen close to 1 and b is chosen close to 0. This completes the proof. □

We conclude the paper with the following remark that our results can be applied to a large variety of bilinear operators between L_p-spaces, in particular to multipliers or to the bilinear Calderón–Zygmund singular operators.

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